

Finals Practice Task 01. Data Preparation and Exploratory Data Analysis (EDA) using Mysql

Description:

This project involved utilizing SQL and MySQL Workbench to effectively clean, organize, and analyze a dataset originating from a call center. The first step was designing a comprehensive database schema to structure the data appropriately, followed by importing the dataset using the Table Data Import Wizard for efficiency and accuracy.

Data cleaning procedures were then carried out, including converting timestamp formats to ensure consistency, correcting invalid or inconsistent scores, and standardizing record entries through various SQL UPDATE statements. To gain insights into the call center operations, exploratory data analysis was conducted by executing multiple SQL queries aimed at identifying peak activity days, assessing service level performance, and calculating key call duration metrics. Advanced techniques such as grouping and window functions were employed to uncover deeper insights into performance trends and patterns.

This process significantly enhanced skills in writing complex SQL queries, managing and maintaining databases, performing data cleaning tasks, and conducting thorough data analysis using MySQL Workbench. Overall, the activity provided a comprehensive experience in harnessing SQL tools for effective data management and analytical insights in a real-world scenario.

The screenshot displays the MySQL Workbench interface. On the left, the 'SCHEMAS' pane shows a project with a table named 'calls'. The 'Query 1' editor in the center contains the following SQL query:

```
72 AVG(call_duration_minutes)
73 FROM calls
74 GROUP BY 1
75 ORDER BY 2 DESC;
76
77 SELECT call_timestamp,
78 MAX(call_duration_minutes) OVER(PARTITION BY call_timestamp) AS max_call_duration
79 FROM calls GROUP BY 1 ORDER BY 2 DESC;
```

Below the query editor, the 'Result Grid' shows the results of the query. The table has two columns: 'call_timestamp' and 'max_call_duration'. The data is as follows:

call_timestamp	max_call_duration
2020-10-04	45
2020-10-22	45
2020-10-26	41
2020-10-13	39
2020-10-21	37
2020-10-03	37
2020-10-14	37
2020-10-30	37
2020-10-09	35
2020-10-12	35
2020-10-25	34
2020-10-16	31
2020-10-19	30
2020-10-02	30
2020-10-20	30

On the far left, the 'Table: calls' structure is shown with the following columns and data types:

- ID: char(10)
- cust_name: char(50)
- sentiment: char(10)
- csat_score: int(11)
- call_timestamp: char(20)
- reason: char(100)
- city: char(50)
- state: char(50)
- channel: char(50)
- response_time: char(10)
- call_duration_minutes: int(11)
- call_center: char(10)